

CSC 445/545 - SUMMER 2026
OPERATIONS RESEARCH: LINEAR PROGRAMMING
ASSIGNMENT 1
UNIVERSITY OF VICTORIA

Due: Sunday, June 14th, 2026 at 11:55pm. Submissions will be accepted without penalty until 7:00am on Monday, June 15th, 2026. **No submissions will be accepted after this time.**

Part 1: LP Solvers [16 marks]

Consider the conjecture below.

In terms of performance, the Simplex Method is not a competitive option for solving LPs in practice.

Your task is to apply your knowledge of linear programming and the Simplex Method to investigate this conjecture and summarize your conclusions. You may use *any methodology you want*, including but not limited to any combination of experimental evaluation, theoretical analysis or a survey of existing research.

Assume that the reader of your report has already taken CSC 445/545; it is not necessary to provide background on aspects of linear programming already covered by the course, and, similarly, it is not sufficient to rely entirely on course material in forming your conclusions. Note that the mark is not influenced by the outcome of your investigation: that is, an answer can receive full marks regardless of whether the conclusions support or refute the conjecture.

If you generate any data or write any code as part of your investigation, please submit that separately; a separate submission entry for supplementary data/code/etc. has been created on Brightspace. If your investigation involves experiments, **all** materials needed to reproduce the experiment must be provided as supplemental materials. You are permitted to use any external sources, including online sources, but you are responsible for choosing high quality sources (and/or cross-referencing if you use less reliable sources). As usual, **you are not permitted to use generative AI in any way** for this assignment. If you consult or use any external sources, include complete citations and a bibliography as part of your written report.

Your report must include a description and justification of your methodology and the results you obtained. It is your responsibility to clearly communicate the work that you did and to show that your conclusions are derived from concrete evidence.

We expect that a typical response will require three typed pages for the written report (not including a bibliography or any diagrams/illustrations/tables, if present), although there is no requirement that the report have any particular length. Your priority should be to present a concise and thorough report of your investigation, rather than to achieve a particular report length.

This task will be marked out of 16 based on the thoroughness of the methodology, justification of the chosen method of investigation, quality of the conclusions, clarity of presentation. A rubric has been posted on Brightspace to provide more detailed guidance on the evaluation process. In general the marking will be based **only** on the information provided in the written report. We may consult your supplemental materials for verification of aspects the written report if necessary, but

we will not mark any of the supplementary materials directly. Please ensure that anything you want considered for marks is contained in your written report.

Part 2: Self Assessment [2 marks]

As noted above, Part 1 will be marked out of 16 based on the thoroughness of the methodology, justification of the chosen method of investigation, quality of the conclusions and clarity of presentation. Your task for Part 2 is to provide an objective and justified self-assessment of your response from Part 1.

A rubric for Part 1 has been posted on Brightspace. Provide a brief but clearly justified self-assessment of your Part 1 response for each of the four categories in the rubric, including a proposed mark for each category. Your proposed mark for each category must be an integer (since the rubric does not support fractional marks). It is your responsibility to clearly justify why your Part 1 response meets the criteria for the proposed mark under each category. When we mark your assignment, we will assess your Part 1 independently before reviewing your self-assessment. If we agree with your self-assessment, we will assign the self-assessed mark.

The mark for Part 2 will be entirely based on the quality of your self assessment, irrespective of the mark for Part 1 (and irrespective of whether we agree with your self assessed mark).

Part 3: Reflection [2 marks]

Your task for Part 3 is to reflect on your learning experience as follows.

- (a) Provide a one paragraph reflection on the process and learning experience of creating your Part 1 response, including a clear description of what original work was performed and what aspects were the most challenging.
- (b) Describe something new that you learned about linear programming while completing this assignment, which has not been covered by the course material (slides/lectures/readings/exercises up to June 10th).